Soru Seti - 3

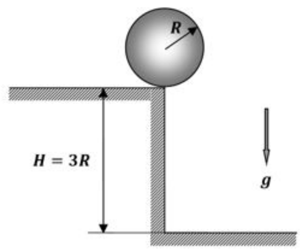
Kutay Kulbak

25 Eylül 2024

1 Mekanik

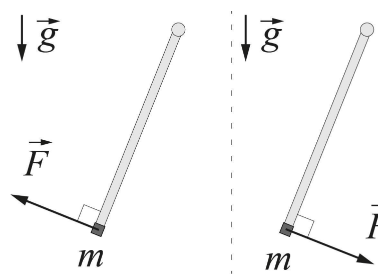
1.1 Düşen Küre

R yarıçaplı m kütleli bir küre $H = 3R$ yüksekliğinde bir duvarın kenarında durmaktadır. Ufak bir itilmeyle küre düşmeye başlıyor. Hareket boyunca küre ve duvarın kenarı arasında kayma olmuyor. Yerçekimi ivmesi g olarak verilmiştir. Hava sürtünmesini ihmal edin. Kürenin v hızıyla kaymadan yuvarlanma yaparken sahip olduğu kinetik enerji $E = \frac{7}{10}mv^2$ olarak veriliyor. Kürenin yere değdiği noktanın duvara uzaklığı l ne kadardır? (Zhautykov 2022, 1.1)



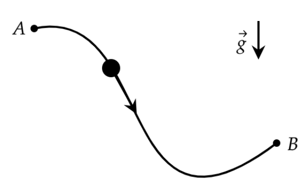
1.2 Rotating Stick

Material point of mass m fixed to the end of a weightless rod that can rotate freely around a horizontal axis O. Initially, the material point is in a position of stable equilibrium under the action of force F, directed perpendicular to the rod (see fig.) At some point the direction of the force F instantly changes to the opposite. Subsequently, the force also remains constant in magnitude and directed perpendicular to the rod, rotating it counterclockwise. At what values F can the rod make a full revolution around the axis of rotation? (Russian Quali, W21, T2)



1.3 Brachistochrone

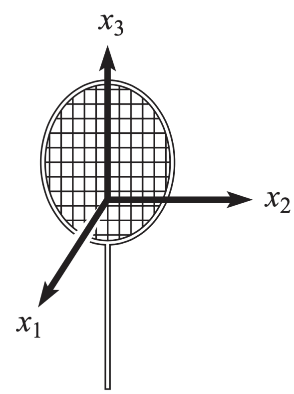
Sabit yerçekimi alanında bir parçacık A noktasından harekete başlıyor. Parçacığın B noktasına varma süresini minimize eden yolu tarif edin. (Johann Bernoulli, 1696)



1.4 Tenis Raketi Teoremi (Dzhanibekov Etkisi)

Bir tenis raketini (ya da kitabı) ana eksenleri etrafında döndürmeye çalışırsanız, farklı eksenler etrafında farklı durumlar yaşandığını göreceksiniz. Kütle merkezine göre farklı ana eksenler etrafındaki eylemsizlik momenti $I_1 > I_2 > I_3$ olarak verilmiş olsun. Raket $\mathbf{\hat{x}_1}$ ve $\mathbf{\hat{x}_3}$ eksenleri etrafında düzgünce dönecek, $\mathbf{\hat{x}_2}$ ekseni etrafında ise kaotik şekilde davranacaktır. Bu durumun sebebini matematiksel olarak açıklayın.
(*İpucu: Raketi bir eksen etrafında döndürmeye çalıştığımızda diğer eksenlerdeki hareketi tamamen engelleyemeyiz.*)

(Morin Mekanik, 9.14)



2 Elektromanyetizma

2.1 Motion of a Charged Ball

A solid, homogeneous spherical ball of mass m and radius R is made of insulating material and has charge Q distributed uniformly throughout its volume. The ball is placed on a large horizontal surface, and set in rolling motion without slipping in such a way that its center starts to move with initial horizontal velocity v_0. There is a uniform magnetic field (flux density) of magnitude B perpendicular to the surface. The coefficient of static friction is large enough to prevent the ball from slipping on the surface. The moment of inertia of the ball about an axis through its center is $2mR^2/5$. Describe the motion of the center of the ball and the shape of its trajectory. (EuPhO 2019, T2)

Hint: Depending on your approach you may use the following identity:

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

2.2 Charged Particle

A particle of mass m and charge q is in a homogeneous magnetic field with induction B (the vector is parallel to the z-axis). The characteristic time of the system is the cyclotron period of the particle $T_B = 2\pi m/Bq$. The system is situated in between two parallel electrodes, which can be used to create an homogeneous, parallel to the x-axis electric field E. (NBPhO 2003, P6)

1) The particle is at rest. At the moment of time $t = 0$, the electric field E is switched on; after a short time interval τ ($\tau \ll T_B$), it is switched off, again. What will be the trajectory of the particle?

2) Let p_x and p_y denote the x- and y components of the momentum of the particle. Sketch the trajectory of the particle in (p_x, p_y)-plane and depict the vectors of the momentum for the moments of time $t_n = nT_B/4$ ($n = 1, 2, 3, 4$).

3) Consider the situation when the on-off switching of the electric field is done periodically, starting with $t = 0$, after equal time intervals $\Delta t = T_B/4$. Sketch the particle trajectories in (p_x, p_y) and (x, y)-planes.

4) Let the period be short, $\Delta t \ll T_B$ (but still much longer than the duration of pulse, $\Delta t \gg \tau$). Show that after the n-th pulse (at the time moment $t_n = n\Delta t$), the momentum of the particle can be represented as the sum of n vectors $\vec{p}_i$, where all the component-vectors are equal in modulus (the modulus being independent of n), and the neighboring vectors ($\vec{p}_i$ and $\vec{p}_{i+1}$, $i = 1, 2, \ldots$) have equal angles between them.

5) Consider the limit case $\Delta t \to 0$, so that $E\tau/\Delta t \to E_k$ (E_k denotes the time-average of the elctric field). Sketch the particles trajectory in (p_x, p_y)-plane and express the particles mean velocity (vectorially; averaged over the cyclotron period) via the quantities E_k and B.

6) Let us return to the non-zero (but still small, $\Delta t \ll T_B$) time-intervals. Let us consider the case when the pulses are of variable polarity: for the $2n$-th pulse, the electric field is $+E$, and for the $2n + 1$st pulse $-E$. Find the particles average velocity, (vectorially) averaged over the cyclotron period.

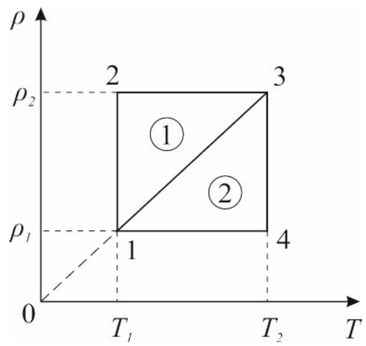
3 Termodinamik

3.1 Çevrim

Verilen (ρ, T) diyagramında, iki ısı makinasının çalıştıkları çevrimler gösterilmiştir. İki ısı makinası da helyum gazı kullanarak çalışmaktadır. Bu iki ısı makinasının verimleri arasındaki ilişki $(3-\eta_1)(1-\eta_2)=1$ olarak verilmiştir. (Rus Takım, X17, T4)

a) Çevrimleri (p, V) diyagramı üstünde gösterin.

b) $T_1 = 50K$ ise T_2'yi hesaplayın.

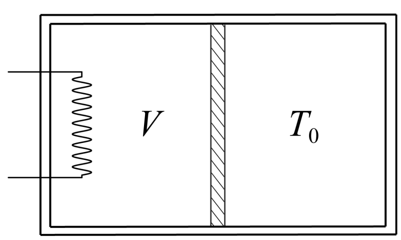


3.2 Cylinder with Heater

A horizontally located cylindrical vessel of volume $2V_0$ is divided into two equal parts by a movable piston and filled with a monoatomic ideal gas. The piston moves without friction. The left wall, the side surface of the entire vessel and the piston do not conduct heat. The temperature of the right wall is maintained constant and equal to the initial temperature of the gas T_0 in both parts of the vessel. The initial gas pressure in both parts is equal to p_0. After turning on the heater, the piston in the left part of the vessel begins to move slowly. (Russian Quali, W21, T3)

a) Determine the dependence of gas pressure p to temperature T on the left side of the vessel.

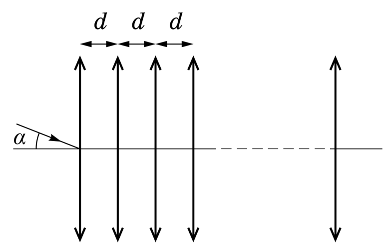
b) For this process, determine how the heat capacity of the gas in the left part of the vessel depends on its volume.



4 Optik

4.1 Close Lenses

The optical system consists of a large number of converging lenses with a focal length F with a common principal optical axis. The centers of adjacent lenses are at the same distance $d \ll F$ from each other, while the thickness of the lens can be neglected in comparison with d. The ray before refraction at the center of the first lens intersects the main optical axis at an angle $\alpha \ll 1$ and immediately after refraction in the latter intersects the main optical axis at the same angle. Find the minimum possible distance L between the outer lenses. (Russian Quali, X21, T2)



4.2 Image of a Circle

In the figure below, the ellipse is a real image of a circle created by an ideal thin lens. The point F is the closer-to-the-ellipse main focus of the lens. Optical axis lies in the plane of this figure. Construct geometrically the centre O of this thin lens. Use the linked website to access the GeoGebra file of the problem. (Physics Cup 2024, P2)